INCH-POUND

MIL-P-71089 (AR)
15 November 1991

MILITARY SPECIFICATION

PROJECTILE, PRACTICE, 155MM, M804E1
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for the Projectile, Practice, 155MM, M804E1 (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

MILITARY

MIL-M-6867	- Magnetic Inspection Units
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specifications For
MIL-P-48395	- Pellet, Explosive Composition (for use in Ammunition)

STANDARDS

MILITARY

MIL-STD-410	- Non-destructive Testing Personnel Qualification and Certification
MIL-STD-1949	- Inspection, Magnetic Particle

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:31Z
Check the source to verify that this is the current version before use.

s and covers) Width: 1½ inches (excluding fittings

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.8)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

12913940	- Spacer
12913926	- Projectile, 155MM: Practice M804E1, Assembly and Marking Diagram
12913935	- Cup

(Copies of drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the Contracting activity or as directed by the Contracting Officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (See 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM-A507	- Steel, Sheet and Strip, Alloy, Hot-Rolled and Cold-Rolled, Drawing Quality

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, NY, NY 10018.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Steel quality of cup. The item contractor shall establish procedures to provide a strip drawing quality alloy steel when procuring steel in accordance with ASTM-A-507, i.e., the steel shall be uniform and free of impurities. The manufacturer will provide a certification to cite the specific revisions of the industry standard under which the steel was manufactured.

3.3 Heat treatment or stress relief. Heat treatment or stress relief applied to the steel cup shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.4 Mechanical properties.

3.4.1 Cup. Mechanical properties specified on the applicable drawing shall apply throughout the component.

3.5 Tensile strength. The cup shall withstand the application of a minimum (min.) tensile force of 15,000 lbs. without breaking.

3.6 Metal defects.

3.6.1 Cup. The cup shall be free from cracks, splits, bursts, folds, seams or other metal defects, as evidenced by magnetic particle inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

3.7 Projectile assembly. The projectile assembly and components shall comply with all requirements specified on Drawing (Dwg.) 12913926 and with all requirements specified in applicable specifications.

3.7.1 Smoke canister. There shall be no foreign matter or smoke composition on the exterior of the canister.

3.8 Proving ground. The projectiles, when tested as specified, shall show no loss or break-up of metal components. The fuzed projectile shall on impact, produce a smoke signature visible across 4000 meters of level terrain.

3.9 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. The requirements for workmanship are as shown on the applicable drawings and specifications and the following:

3.10.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and specification. Cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.10.2 Protective coating. Protective coating shall comply with the requirements of the applicable drawings and specifications. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.10.3 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determinations.

3.10.4 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.10.5 Burr. No part shall have a burr which might interfere with the assembly or function of the projectile or which might be injurious to personnel handling the item.

3.10.6 Foreign matter. No part or assembly shall contain dirt, grease, rust, corrosion or other foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

3.10.7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

Part Description	Drawing	Quantity
Projectile, 155MM, Practice M804E1 Assembly and Marking Diagram	12913926	25
Cup	12913935	5
Spacer	12913940	5
Disc Closing	12913932	5
Pad	12913933	5
Pellet	12913937	5
Explosive Cup	12913939	5

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assemblies and Components	1 OF 2	
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Projectile, 155MM, Practice M804E1 Assembly and Marking Diagram (Dwg. 12913926)			
	Examination for Defects (Except for Proving Ground Test)	5, Level I	3.7	4.4.2
	Proving Ground Test	20, Level I	3.8	4.5.7
	Components			
	Cup (Dwg. 12913935)			
	Examination for Defects	5, Level I	3.7	4.4.2.1
	Spacer (Dwg. 12913940)			
	Examination for Defects	5, Level I	3.7	4.4.2.8

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
	Projectile Assemblies and Components		NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Disc Closing (Dwg. 12913932) Examination for Defects	5, Level I	3.7	4.4.2.2
	Pad (Dwg. 12913933) Examination for Defects	5, Level I	3.7	4.4.2.3
	Pellet (Dwg. 12913937) Examination for Defects	5, Level I	3.7	4.4.2.4
	Explosive Cup (Dwg. 12913939) Examination for Defects	5, Level I	3.7	4.4.2.5

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain loaded projectile of only one weight zone. The following inspection lot formation shall apply for cup, steel (dwg 12913935):

a. Heat treatment lots. Heat treatment lots shall contain heat treated cups from not more than one heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the hardness test hereinafter prescribed.

b. Stress relieved lots. Stress relieved lots shall contain stress relieved cups, from not more than one heat of steel, that have been cold shaped by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the hardness test hereinafter prescribed.

c. Miscellaneous lots. Small lots (such as cups manufactured from abnormally small heats of steel or cups that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either (a) or (b) above, as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the hardness test hereinafter prescribed.

d. Cup lots. When it has been determined that cups of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be grouped for further processing.

4.4.1.1 Shipping lots. Shipping lots shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawings, revision, under conditions of a continuous production, free of interruptions other than those due to the end of shift day, or work week. Shipping lots shall be assigned a lot number in accordance with MIL-STD-1168. The shipping lots thus constituted shall be the basis for selection of samples for the proving ground functional test. Lot size shall be determined by the procuring contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.2 Examinations and tests.

a. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristic paragraphs.

TABLE II Attributes Sampling Inspection

Lot Size	Inspection Level					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	30	20	8
281 to 500	*	125	32	30	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.10). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-71089 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup, Steel	1 OF 2	12913935
			NEXT HIGHER ASSEMBLY 12913926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Magnetic Particle Test	Note 1	3.6.1	4.5.2
102	Magnetic Particle Test Stamp Missing	Ø-1, Level I	3.6.1	Visual
103	Rockwell Hardness	Note 2	3.4.1	4.5.1.1
104	Tensile Force	20, Ø-1	3.5	4.5.6
105	Circular Runout of I.D.	Ø-1, Level II	3.7	Gage
106	I.D. max	Ø-1, Level II	3.7	Gage
107	Wall Thickness, min	Ø-1, Level II	3.7	Gage
108	Thickness of Base min	Ø-1, Level II	3.7	Gage
109	Thread Pitch Diameter, min.	Ø-1, Level II	3.7	Gage
110	Thread Major Diameter, min.	Ø-1, Level II	3.7	Gage
111	Length, max.	Ø-1, Level II	3.7	Gage
112	Length of Thread, min.	Ø-1, Level II	3.7	Gage

NOTES:
Note 1: Sampling in accordance with 4.4.3.3
Note 2: Sampling in accordance with 4.4.3.2.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-71089 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup, Steel	2 OF 2	12913935
			NEXT HIGHER ASSEMBLY 12913926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Any Feature Missing	0-1, Level IV	3.7	Visual
202	Surface Finish Improper	0-1, Level IV	3.10.3	Visual-ANSI-B46.1
203	Protective Coating Inadequate or Damaged	0-1, Level IV	3.10.2	Visual
204	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Disc, Closing	1 OF 1	12913932
			NEXT HIGHER ASSEMBLY 12913936

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Crack, Split, Puncture or Similar Damage Extending Through Material	0-1, Level II	3.7	Visual
102	Thickness	0-1, Level II	3.7	Gage
Minor				
201	Diameter	0-1, Level IV	3.7	Gage
202	Damage (not extending through material) or distortion	0-1, Level IV	3.7	Visual
203	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Pad	1 OF 1	12913933
			NEXT HIGHER ASSEMBLY 12913936

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Thickness	0-1, Level III	3.7	Gage
Minor				
201	Diameter	0-1, Level IV	3.7	Gage
202	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Pellet	1 OF 1	12913937
			NEXT HIGHER ASSEMBLY 12913926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Cracked, Chipped or Broken	Ø-1, Level II	3.7	Visual
102	Weight	Ø-1, Level II	3.7	Balance
103	Diameter	Ø-1, Level II	3.7	Gage
104	Height	Ø-1, Level II	3.7	Gage
105	Moisture Content of Explosive Charge Material	Note 1	3.7	4.5.4
106	Density of Explosive Charge Material	Note 2	3.7	4.5.5
Minor				
201	Radii max.	Ø-1, Level V	3.7	Gage
202	Evidence of Poor Workmanship	Ø-1, Level V	3.10	Visual

NOTES: Note 1. Sampling in accordance with 4.5.4
Note 2. Sampling in accordance with 4.5.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-71089 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Explosive Cup	1 OF 1	12913939
			NEXT HIGHER ASSEMBLY 12913936

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Crack, Split, Puncture or Similar Damage Extending Through Material	0-1, Level II	3.7	Visual
102	Thickness Through Base And Sidewall min.	0-1, Level II	3.7	Gage
103	Height	0-1, Level II	3.7	Gage
104	Inside Diameter	0-1, Level II	3.7	Gage
Minor				
201	Radii Incorrect	0-1, Level IV	3.7	Gage
202	Damage (not extending through Material) or Distortion	0-1, Level IV	3.7	Visual
203	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Pellet Assembly (prior to crimping of closing disc assy)	1 OF 1	12913936
			NEXT HIGHER ASSEMBLY 12913926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Charge Pellet Stack Sequence (felt pad, pellet, felt pad, closing disc)	0-1, Level II	3.7	Visual
102	Cup Cracked or Damaged	0-1, Level II	3.7	Visual
103	Pellet Cracked or Damaged	0-1, Level II	3.7	Visual
Minor				
201	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-71089 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Pellet Assembly	1 OF 1	12913936
			NEXT HIGHER ASSEMBLY 12913926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Pellet Loose	Ø-1, Level I	3.7	Manual
102	Length max.	Ø-1, Level II	3.7	Gage
103	Outside Diameter max.	Ø-1, Level II	3.7	Gage
104	Closing Disc Bulge Exceeding max. Length of Pellet Assy.	Ø-1, Level II	3.7	Gage
105	Closing Disc Damaged	Ø-1, Level II	3.7	Visual
106	Cracks in Aluminum Crimp Area	Ø-1, Level II	3.7	Visual
Minor				
201	Tab Missing, Incorrect or Inadequate	Ø-1, Level IV	3.7	Visual
202	Crimp Not Full 360 Degrees	Ø-1, Level IV	3.7	Visual
203	Wrinkled Cup	Ø-1, Level IV	3.7	Visual
204	Marking Missing, Misleading or Unidentifiable	Ø-1, Level IV	3.7	Visual
205	Explosive on Outside Surface	Ø-1, Level IV	3.10	Visual
206	Evidence of Poor Workmanship	Ø-1, Level IV	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Spacer	1 OF 1	12913940
			NEXT HIGHER ASSEMBLY 12913926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Height	0-1, Level III	3.7	Gage
102	Outside Diameter min.	0-1, Level III	3.7	Gage
103	Inside Diameter max.	0-1, Level III	3.7	Gage
Minor				
201	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

AMS Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Projectile and Cup Assembly Prior to Installation of Smoke Canister	1 OF 1	12913926
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Depth of Cup, min.	0-1, Level II	3.7	Gage
102	Depth of Cup, max.	0-1, Level II	3.7	Gage
103	Presence of Cup Thread Adhesive, Missing, Incorrect or Inadequate	0-1, Level II	3.7	Visual/witness during assembly
104	Torque Cup (witness during application of assembly torque)	0-1, Level II	3.7	Visual/witness during assembly
Minor				
201	Corrosion Preventive Compound Missing, Incorrect or Inadequate on Threads	0-1, Level IV	3.7	Visual
202	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Projectile, 155MM, Practice M804E1 Loading Assembly and Marking Diagram	1 OF 2	12913926
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Diameter of Bourrelet, max (applies if bourrelet has been retouched)	100%	3.7	Gage
102	Projectile Zone Weight Incorrect	100%	3.7	Balance
103	Zone Stenciling Incorrect	100%	3.7	Visual
104	Any Weight Zone Punch Mark Inadequate	0-1, Level II	3.7	Visual/Manual
105	Smoke Canister or pellet assembly Missing	100%	3.7	Visual
106	Spacer Missing	0-1, Level II	3.7	Visual
107	Lifting Plug Gasket Missing	0-1, Level II	3.7	Visual
108	Lifting Plug Missing	0-1, Level II	3.7	Visual
109	Torque Test Failure of Lifting Plug	0-1, Level II	3.7	Test/4.5.3
Minor				
201	Corrosion Preventive Compound not completely Covering Fuze Threads	0-1, Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Projectile, 155MM, Practice M804E1 Loading Assembly and Marking Diagram	2 OF 2	12913926
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
202	Plug not Full Seated	0-1, Level IV	3.7	Visual
203	Rotating Band Damaged	0-1, Level IV	3.7	Visual
204	Grommet Missing, Damaged or Improperly Assembled	0-1, Level IV	3.7	Visual
205	Protective Coating with Total Damaged Area in Excess of 645 Square MM (1 square inch)	0-1, Level IV	3.7	Visual
206	Marking Misleading or Unidentifiable (except for zones) Including Incorrect Color Code	0-1, Level IV	3.7	Visual
207	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Packaging and Marking	1 OF 1	9331807
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Pallet Cover Damaged or Improperly Assembled	0-1, Level II	3.7	Visual
102	National Stock Number or DoD Code Missing or Unidentifiable	0-1, Level II	3.7	Visual
103	Projectile of Different Zones on the Same Pallet	0-1, Level II	3.7	Visual
104	Strapping or Wire Missing or Loose	0-1, Level II	3.7	Manual
Minor				
201	Number of Projectile Assemblies Incorrect	0-1, Level IV	3.7	Visual
202	Marking Misleading or Unidentifiable	0-1, Level IV	3.7	Visual
203	Strapping or Wire Improperly Engaged	0-1, Level IV	3.7	Visual
204	Evidence of Poor Workmanship	0-1, Level IV	3.10	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Steel quality. Records of cup, steel test documentation shall be maintained and made available for Government review to provide evidence of compliance with the requirements of 3.2.

4.4.3.2 Mechanical properties.

4.4.3.2.1 Hardness.

4.4.3.2.1.1 Heat treatment lots. In order to determine uniformity of heat treatment, each cup in the heat treatment lot (as defined by 4.4.1a) after heat treatment, shall be subjected to a hardness test with a Rockwell Hardness Tester or any other method approved by the procuring activity. Hardness impressions shall be made on the cup sidewall.

4.4.3.2.1.2 Stress relieved lots. Twenty-five (25) cups from the first batch of each stress relieve lot (as defined by 4.4.1b) processed through the stress relief furnace and 5 cups from each one hours' production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the product.

4.4.3.2.1.3 Miscellaneous lots. Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.4.3.2.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness test in accordance with 4.4.3.2.1.2.

4.4.3.3 Magnetic particle inspection. Major defect, heat treated or stress relief cups after final machining shall be 100 percent inspected by the magnetic particle inspection technique approved by the responsible technical agency.

a. Should a controversy arise as to the validity of an indication, the manufacturer shall use metallographic test methods or an approved non-destructive method for proof of acceptability to the responsible technical agency.

b. Magnetic particle inspection shall not be used as an alternative procedure in lieu of corrective manufacturing process action to eliminate a continuing process induced defect. If corrective action is not instituted and the cause eliminated, the Government reserves the right to withdraw acceptance inspection until the contractor can prove his process is capable of producing suspect free cups.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Inspection Method Reference" column in the table starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Mechanical properties.

4.5.1.1 Hardness (cup). The test shall be made using the procedures as prescribed in ASTM-E18. The readings shall be taken half way down the cup sidewall. Hardness tester periodic verification shall be performed using Rockwell test standards manufactured in accordance with ASTM method E-18, in lieu of production samples.

4.5.2 Magnetic particle (Inspection method). The procedure for performing this test utilizing the wet fluorescent method shall be as specified in MIL-STD-1949 for indirect magnetization by either the continuous method or the residual method utilizing longitudinal and circular magnetic fields. The suspension shall be applied to the entire cup, inside and outside. After draining, the cup shall be thoroughly inspected in a darkened area under an ultra-violet light which will produce adequate fluorescence of the magnetic particles. The solid content of the suspension shall be maintained at a level recommended by the suspension manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by specification MIL-STD-1949. When performing the test for solid content of the suspension, if it is determined that it does not comply, all cups that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. Magnetic particle inspection shall be performed on equipment that complies with the requirements of MIL-M-6867. If any cup fails to meet the requirements of 3.6.1, that cup shall be removed from the lot.

4.5.2.1 Operator qualifications. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD- 410, Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.5.3 Torque test. This test shall be performed on the sample selected in accordance with 4.4.2.10. A thin line shall be drawn across the joint in question to aid in observing movement. The minimum and maximum disassembly torque for the lifting plug shall be applied in the disassembly direction. The plug shall loosen at the max. disassembly torque and shall not loosen at the min. disassembly torque.

4.5.4 Moisture content of explosive charge material for pellet (dwg. 12913937). When the explosive materiel is suspected of having been subjected to adverse storage or shipping conditions permitting the possibility of moisture contamination, a determination of moisture content shall be made. A representative sample of each material shall be selected and tested in accordance with MIL-P-48395. If the moisture content fails to comply with the requirement specified in the applicable specification, and loading has not begun, the explosive represented by the sample shall not be used for production until corrective action has been taken as verified by repeating this test. In addition, if loading has begun, pellet assemblies with the non-conforming composition shall be rejected.

4.5.5 Consolidation density of explosive composition of pellet (dwg. 12913937). The density of the explosive composition of the pellet shall be determined for each consolidate tool at the start of production each day and after every two hours of production or portion thereof. If the density of any pellet in a representative test sample of three (3) fails to comply with the applicable drawing requirement, the pellets produced since the last satisfactory check shall be rejected. In addition, the necessary adjustments shall be made to the consolidate tool which will be verified by determining the density of three (3) consecutively produced pellets.

The density of the explosive composition of the pellet shall be calculated as follows:

$$\text{Density, gm/cc} = \frac{\text{weight in gms}}{(\text{Height, inches})\ (\text{Diameter, inches})^2\ 12.870}$$

4.5.6 Tensile test of cup. The tensile test of the cup shall be performed as follows:

Thread the cup into a threaded stationary fixture applying the required seating torque. The fixture internal cross section shall be identical to the projectile threaded nose area. Apply the tensile force to the bottom of the cup with a ram-rod slightly smaller in diameter than the cup inside diameter. The base of the ram-rod shall have a slightly larger radii than the cup. The outside diameter of the ram-rod shall be necked down to 1.700 inches max. (so as not to support the cup in the threaded area) for twice the total cup threaded length. The applicable force shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

IOS ... n that issue of the Department of Index of Specifications and Standa

applied gradually until the min. tensile force is reached. If any cup breaks, the entire lot shall be rejected. After testing, all undamaged cups that meet the inspection criteria of para 4.4.2.1 may be returned to the lot.

4.5.7 Proving ground. These tests shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. The sample projectiles selected shall be fired in a M185 or M199 Series cannon using a M119 series, Zone 8 charge and a M557 or M739 series fuze set "super quick". Chamber pressures will be based on copper pressure gages. The projectiles and propelling charges shall be uniformly temperature conditioned to 21° Celsius (70°F) plus or minus 1° Celsius (2°F) for a minimum of 24 hours prior to testing. Adequate photographic and instrumental coverage shall be provided to assure compliance with requirements. Range and deflections shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts for informational purposes. An observer shall be stationed 4000 ± 200 meters from the edge of the impact area to record presence of smoke signature. The test shall be conducted on level terrain and the wind velocity shall not exceed 1' knots at the time of testing.

5. PACKAGING

5.1 Packing and marking. The projectile shall be packed and marked in accordance with provisional packing and marking instructions to be supplied by the Government procuring agency.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The projectile covered by this specification is intended for use as a target practice round for 155MM Howitzers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Applicable stock number.

c. First article sample requirments.

d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (See 2.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

6.3 Submission of contractor inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to: Commander, U.S. Army Armament, Munitions and Chemical Command, ATTN: SMCAR-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.6 Submission of test data and ammunition data cards. In addition to the normal distribution of records, when the projectile is procured by the U.S. Army Armament, Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARDEC, ATTN: SMCAR-CCH-P and SMCAR-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000, and Commander, ARRCOM, ATTN: AMSMC-QAD, Rock Island, IL 61299-6000.

6.7 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

6.8 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research, DPvelopment and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, U.S. Army Research and Development Command (ARRADCOM) or U.S. Army Armament Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.9 Heat treatment. Heat treatment is understood to be any thermal process which brings about a modification in the mechanical and metallurgical properties of the metal involved.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

6.10 Submission of alternative quality conformance provision. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.11 Subject term (key word) listing.

Ammunition
Heat treatment
Magnetic particle inspection
Moisture content
Tensile strength

Custodian
Army - AR

Preparing activity:
Army - AR

(Project 1320-AC67)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisi letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-71089 (AR)	15 NOVEMBER 1991

3. DOCUMENT TITLE
PROJECTILE, PRACTICE, 155MM, M804E1 LOADING, ASSEMBLING AND PACKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6675	(2) AUTOVON 880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.